Species-Aware Module Review: Bacterial Glycogen / α-Glucan Synthesis and Mobilization in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI txid 160488, proteome UP000000556) **Module / bucket:** kegg:ppu00500 "Starch and sucrose metabolism" (generic storage-carbon module: branched glycogen/α-glucan synthesis + mobilization) **Evidence basis:** Direct genome/proteome enumeration (UniProt UP000000556, InterPro/Pfam) for the target strain; mechanistic transfer from *Mycobacterium*/*Streptomyces* GlgE-pathway biochemistry (species transfer flagged throughout).

---

1. Executive summary

The generic module assumes glycogen is built by a nucleotide-sugar glycogen synthase (canonical **GlgC/ADP-glucose → GlgA**, or a proposed **GalU/UDP-glucose → GlgA**) and branched by GlgB, then mobilized by GlgP + GlgX. **This framing is wrong for *P. putida* KT2440.**

Direct proteome evidence shows:

- **`glgC` (ADP-glucose pyrophosphorylase, EC 2.7.7.27) is entirely absent** (0 hits by gene name, EC, and protein name). The canonical ADP-glucose route therefore **cannot operate** → `not_expected_in_target_taxon`.
- The only annotated glycogen synthase, **PP_4050 `glgA` (EC 2.4.1.21)**, carries an **ADP-glucose-type GT5 / starch-synthase family signature** (IPR011835, IPR013534/PF08323) but has **no ADP-glucose donor and no UDP-glucose-synthase alternative** (EC 2.4.1.11 absent). It is a **substrate-orphan** → `candidate_uncertain`; **promote to full review**.
- The organism instead encodes a **complete actinobacterial-type GlgE (TreS–Mak–GlgE–GlgB) α-glucan pathway** (PP_4059 TreS-Mak fusion, PP_4060 GlgE, PP_4058 GlgB) plus **TreY–TreZ** (PP_4053/PP_4051) and **MalQ** (PP_4052). This is the operative synthetic arm and is the glycogen-like α-glucan polymer source in this strain.

- Mobilization is cleanly covered by a **single GlgP** (PP_5041, EC 2.4.1.1; no separate MalP) and a **family-confident GlgX** (PP_4055, IPR011837).
- Trehalose that feeds the GlgE pathway is made by **TreY–TreZ** + **TreS**, **not OtsAB** (OtsA/OtsB absent), so the "glycogen", "α-glucan (GlgE)" and "trehalose" modules are a single interconnected network here and cannot be cleanly separated as the generic boundaries assume → `module_needs_revision`.

Bottom line: In *P. putida* KT2440 the storage α-glucan is a **GlgE-pathway product from trehalose/maltose-1-phosphate, not a GlgC/ADP-glucose glycogen**. Curate the ADP-glucose route as a gap, the UDP-glucose route as unsupported/uncertain, and add the TreS–Mak–GlgE + TreY–TreZ steps into the operative module.

---

2. Target-organism pathway definition

Included biochemical process (as realized in KT2440): Cytoplasmic synthesis of a branched α-1,4/α-1,6-glucan (glycogen-like) from **maltose-1-phosphate** via the maltosyltransferase **GlgE**, with branching by **GlgB**; and its mobilization back to glucose-1-phosphate via **GlgP** (phosphorolysis) and **GlgX** (α-1,6 debranching), with **MalQ** (amylomaltase) redistributing maltooligosaccharides. The α-glucan is metabolically continuous with trehalose through **TreS/Mak** (trehalose↔maltose→maltose-1-P) and **TreY–TreZ** (α-glucan/maltooligosaccharide→trehalose). Module endpoint is glucose-1-phosphate.

Neighboring processes to keep separate (boundaries): - **Central carbon interconversion downstream of G1P** — `pgm` (PP_3578) and `algC` (PP_5288) phosphoglucomutases, and `pgi1`/`pgi2` glucose-6-P isomerases (PP_1808/PP_4701). These are glycolysis/gluconeogenesis, **not** storage-glucan steps; they are boundary/downstream, not module members. - **Cellulose biosynthesis** — `bcsA` (PP_2635, cellulose synthase, EC 2.4.1.12) makes β-1,4-glucan; **wrong polymer**, exclude from this α-glucan module despite its ppu00500 bucket tag. - **Periplasmic β-glucoside catabolism** — `bglX` (PP_1403, β-glucosidase, EC 3.2.1.21); unrelated hydrolysis, exclude. - **Free-glucose phosphorylation / mannose branch** — `glk` (PP_1011), `cpsG` (PP_1777 phosphomannomutase); peripheral, exclude. - **Nucleotide-sugar supply for LPS/EPS** — `galU` (PP_3821, UDP-glucose pyrophosphorylase); UDP-glucose primarily serves LPS/exopolysaccharide, not α-glucan (see §5).

Alternate names / database definitions: KEGG ppu00500 "Starch and sucrose metabolism" is a broad overview map that lumps glycogen, trehalose, maltose, cellulose and sucrose reactions; it should not be equated with a mechanistic "glycogen module". The operative route is the **GlgE / TreS–Mak–GlgE pathway** (a.k.a. the trehalose→α-glucan pathway; MetaCyc "glycogen biosynthesis II / GlgE pathway"). "Glycogen" and "α-glucan/capsular glucan" are used interchangeably in the GlgE literature.

---

3. Expected step model (generic step → status in KT2440)

Generic module step	Status in *P. putida* KT2440	Basis
GlgC-dependent ADP-glucose formation	**not_expected_in_target_taxon (gap)**	No EC 2.7.7.27 / `glgC` in proteome (direct)
ADP-glucose-dependent GlgA chain extension	**gap / likely over-annotation**	Only PP_4050 GlgA present but no ADP-glucose donor (direct)
GalU-dependent UDP-glucose formation	**covered (but for LPS/EPS, not glucan)**	PP_3821 GalU present (direct); role assignment uncertain
UDP-glucose-dependent Pseudomonas GlgA extension	**candidate_uncertain**	No dedicated UDP-glucan synthase (EC 2.4.1.11 absent); PP_4050 family is ADP-glucose-type (direct)
GlgB α-1,6 branching	**covered**	PP_4058 GlgB, IPR006407 (direct)
GlgP phosphorolysis	**covered**	PP_5041 GlgP, sole EC 2.4.1.1 (direct); glycogen vs maltodextrin substrate caveat
GlgX debranching	**covered**	PP_4055 GlgX, IPR011837 (direct)
(missing from generic model) TreS–Mak → maltose-1-P	**covered; should be ADDED to module**	PP_4059 TreS-Mak fusion (direct)
(missing from generic model) GlgE maltosyltransferase (α-glucan synthesis)	**covered; the OPERATIVE synthesis step**	PP_4060 GlgE, EC 2.4.99.16 (direct)
(missing) TreY–TreZ trehalose formation; MalQ	**covered; interlinked**	PP_4053/PP_4051/PP_4052 (direct)

4. Candidate genes and evidence

High-confidence, module-core (direct proteome + family evidence): - **PP_4060** `glgE` — α-1,4-glucan:maltose-1-P maltosyltransferase (EC 2.4.99.16; GH13 PF00128 + IPR026585 GlgE). **Operative α-glucan synthase.** Confident. - **PP_4059** `treSB` — bifunctional **TreS–maltokinase (Mak/Pep2) fusion** (EC 5.4.99.16 + EC 2.7.1.175; 1106 aa; TreS + maltokinase InterPro domains). Supplies maltose-1-P. Confident; note the dual EC/fusion when mapping GO. - **PP_4058** `glgB` — 1,4-α-glucan branching enzyme (EC 2.4.1.18). Confident. - **PP_4055** `glgX` — glycogen/limit-dextrin debranching enzyme (EC 3.2.1.33; IPR011837 GlgX). Confident. - **PP_5041** `glgP` — α-1,4-glucan phosphorylase (EC 2.4.1.1; IPR011833/IPR000811). Confident for phosphorolysis; **caveat:** the GlgP vs MalP (maltodextrin phosphorylase) physiological distinction is not resolvable from sequence alone — single-copy enzyme likely serves both. - **PP_4052** `malQ` — 4-α-glucanotransferase/amylomaltase (EC 2.4.1.25). Confident; maltooligosaccharide remodeling. - **PP_4053** `treY` / **PP_4051** `treZ` — maltooligosyltrehalose synthase/trehalohydrolase (EC 5.4.99.15 / 3.2.1.141). Confident; trehalose formation from α-glucan (interlinked module).

Ambiguous / caveated: - **PP_4050** `glgA` — annotated glycogen synthase (EC 2.4.1.21), ADP-glucose-type GT5 family (IPR011835). **Orphan: no ADP-glucose donor; no UDP-glucan-synthase paralog.** Roles possible: (a) vestigial/substrate-starved; (b) non-canonical donor (UDP-glucose) — unproven; (c) maltose-1-P-forming activity feeding GlgE (by analogy to *M. tuberculosis* GlgA, P 27513637) — but that too needs ADP-glucose. **candidate_uncertain; promote to full review.** - **PP_3821** `galU` — UDP-glucose pyrophosphorylase (EC 2.7.7.9). Present, but UDP-glucose in *Pseudomonas* is principally for LPS/EPS; its role as a glucan-synthesis donor is **inferential and weak** without a UDP-dependent synthase. - **PP_2918** `treSA` — standalone trehalose synthase (EC 5.4.99.16); paralog of the TreS moiety of PP_4059; relevant to trehalose↔maltose interconversion, not directly α-glucan.

Off-target / boundary candidates (exclude from module): `bcsA` (β-glucan/cellulose), `bglX` (β-glucosidase), `glk`, `cpsG`, `pgi1`, `pgi2`, `pgm`, `algC` — central-carbon or unrelated-polymer genes bucketed into ppu00500 by the broad KEGG map.

5. Gaps, ambiguities, and likely over-annotations

1. **`glgC` gap (high confidence, direct, two databases):** No ADP-glucose pyrophosphorylase → the canonical glycogen route is genomically absent. **Independently confirmed in KEGG:** ortholog K00975 (glgC) returns no `ppu` gene, while PP_4050 maps to K00703 (ADP-glucose glycogen synthase), PP_4060→K16147 (glgE), PP_5041→K00688 (glgP). Standalone maltokinase KO K16149 also has no `ppu` gene, consistent with the Mak activity being fused into PP_4059. Contrast: other proteobacteria (e.g., *Rhodobacter sphaeroides*) carry a full `glgCAPXB` operon (P 10729189), so absence in *P. putida* is a real lineage difference, not a database artifact.
2. **PP_4050 `glgA` = likely over-propagated "glycogen synthase":** EC 2.4.1.21 implies an ADP-glucose glycogen synthase that has no substrate here. The annotation should be softened (e.g., "putative glycogen/α-glucan synthase, donor uncertain") pending experimental donor determination.
3. **"Pseudomonas GalU/UDP-glucose GlgA route" (generic module hypothesis) is unsupported by the genome:** there is no EC 2.4.1.11 UDP-glucose-dependent glucan synthase, and the sole GlgA is ADP-glucose-family-typed. Mark this module branch **candidate_uncertain / speculative** for KT2440.
4. **GlgP substrate ambiguity:** GlgP/MalP cannot be distinguished by annotation; single-copy PP_5041 likely acts on both glycogen and maltodextrins.
5. **Module-boundary error:** GlgE, TreS–Mak and TreY–TreZ are treated as "neighboring modules" generically, but in KT2440 they are the **only** functional α-glucan synthesis/interconversion route and must be folded in.

---

6. Module and GO-curation recommendations

- **Mark `covered`:** GlgB (PP_4058), GlgX (PP_4055), GlgP (PP_5041), plus the GlgE-pathway steps GlgE (PP_4060), TreS-Mak (PP_4059), TreY/TreZ (PP_4053/PP_4051), MalQ (PP_4052).
- **Mark `not_expected_in_target_taxon` / `gap`:** GlgC-dependent ADP-glucose formation and the ADP-glucose-dependent GlgA extension step.
- **Mark `candidate_uncertain`:** the UDP-glucose/GlgA branch (donor unproven) and PP_4050 GlgA itself.

- **Mark `module_needs_revision`:** promote **TreS–Mak–GlgE α-glucan synthesis** and **TreY–TreZ** from "neighboring" to **in-module** for *Pseudomonas*/KT2440; add maltose-1-phosphate as an explicit intermediate. Consider a distinct module document **"Glycogen/α-glucan biosynthesis via the GlgE (trehalose→maltose-1-P) route"** for glgC-negative bacteria.
- **Cross-database coverage confirmation:** every "covered" gene is confirmed in both UniProt and KEGG — glgB K00700→PP_4058, glgX K01214→PP_4055, treY K06044→PP_4053, treZ K01236→PP_4051, malQ K00705→PP_4052, treS K05343→PP_4059, glgE K16147→PP_4060, glgP K00688→PP_5041. **Caveat:** the standalone maltokinase/Pep2 KOs (K16055, K16149) have no `ppu` gene, so the maltose-1-phosphate-forming Mak step is invisible in KEGG's KO scheme; it must be credited to the **PP_4059 TreS-Mak fusion** (EC 2.7.1.175) and **not** marked a gap on the basis of KEGG KO absence.
- **GO considerations:** GlgE activity = GO:0102500 (maltose-1-phospho-...maltosyltransferase-type); ensure PP_4060 is annotated to the GlgE/maltosyltransferase term, not to generic glycogen synthase (GO:0004373). Flag PP_4050 for review of GO:0004373 (glycogen [starch] synthase) given missing donor — note its KEGG assignment K00703 (ADP-glucose glycogen synthase) is the source of the over-propagated ADP-glucose activity. Maltokinase GO:0043915 for PP_4059's Mak activity.

---

7. Genes to promote to full `fetch-gene` review

1. **PP_4050 `glgA` (Q88FN9)** — highest priority: resolve donor specificity (ADP- vs UDP-glucose), functionality, and whether it feeds GlgE; its annotation drives the module's synthetic-arm decision.
2. **PP_5041 `glgP` (Q88CY8)** — resolve GlgP vs MalP physiological substrate.
3. **PP_3821 `galU` (Q88GA4)** — confirm whether UDP-glucose has any glucan-synthesis role vs LPS/EPS only.
4. **PP_4059 `treSB` (Q88FN0)** — verify the TreS+maltokinase bifunctionality and correct dual-EC/GO mapping.

---

8. Key references

- Kalscheuer R *et al.* 2010,

  P 20305657

  — TreS–Pep2(Mak)–GlgE–GlgB pathway from trehalose to α-glucan (*M. tuberculosis*); defines the operative route (species transfer: mechanism strong, genes homologous to KT2440 PP_4059/PP_4060/PP_4058).
- Koliwer-Brandl H *et al.* 2016,

  P 27513637

  — α-glucan made **exclusively** via maltose-1-P/GlgE; GlgA's classical glycogen-synthase activity is weak; convergence of TreS-Pep2 and GlgC-GlgA routes.
- Miah F *et al.* 2016,

  P 27121970

  — *Streptomyces venezuelae* has the GlgE pathway but **lacks glgC and glgA**; GlgE necessary and sufficient for α-glucan — the closest genomic analogue of the KT2440 configuration.
- Roy R *et al.* 2013,

  P 23901909

  — TreS–Pep2 hetero-octameric complex regulating maltose-1-P flux.
- Mendes V *et al.* 2015,

  P 26616850

  ;

  Kermani AA *et al.* 2019,

  P 30877199

  — GlgE and TreS:Pep2 structures.
- Igarashi RY & Meyer CR 2000,

  P 10729189

  — canonical `glgCAPXB` operon and ADP-glucose pyrophosphorylase in the proteobacterium *Rhodobacter sphaeroides* (contrast: canonical route present elsewhere, absent in *P. putida*).

Direct vs inferred: Gene presence/absence statements (glgC, OtsAB, EC counts, family signatures) are **direct** for UP000000556. Pathway *function* (that α-glucan is made by GlgE and not GlgA) is **inferred** from homology + the cited GlgE-pathway biochemistry in Actinobacteria; species transfer to *Pseudomonas* is **moderate-to-strong** at the genomic level but lacks a direct KT2440 knockout/metabolite study — the key open experiment.
